

Symposium on Probability and Entropy in Physics and Beyond

JUNE 13

University at Albany,
Main Campus
Lecture Center 19

A symposium on the role of Bayesian and entropic methods of inference in the natural sciences, ranging from biological systems to the foundations of quantum mechanics

PROGRAM DETAILS



pessoap.github.io/symposium

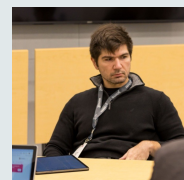
Co-organizer

Dr. Pedro Pessoa
Arizona State University



Co-organizer

Dr. Selman Ipek
Leibniz Universität Hannover



University at Albany,
Department of Physics,
1400 Washington Ave. 12222



Symposium on Probability and Entropy in Physics and Beyond

Abstract Book

University at Albany
June 13, 2026

Watching online

A Zoom connection will be available for those who would like to watch the symposium talks online.

Meeting ID: **876 6248 9183** ([Link](#))

Passcode: **12345**

About the Symposium

The goal of the symposium is to bring together researchers from various disciplines that are unified in their application of the Bayesian and entropic methods to solve fundamental problems in the sciences. Being of universal applicability, such methods find uses in domains ranging from biological systems to the foundations of quantum mechanics. We hope the symposium encourages a lively intellectual discussion between researchers in these distinct domains.

The work of Ariel Caticha has been crucial to the development of our modern understanding of Bayesian and entropic methods of inference as well as their role in the foundations of physics. This symposium serves an additional role as commemorating his many contributions to this field and the people involved with it.

Code of Conduct

We expect all participants in the symposium to:

- Exercise respect in your speech and actions.
- Refrain from demeaning, discriminatory, or harassing behavior and speech.
- Abstain from making use of sexual & sexualized imagery in talks and posters.
- Be mindful of your surroundings and your fellow participants: for example, other people may hear inappropriate comments even if they are not your intended audience.
- Alert community leaders and get involved (if safe and possible) when you notice a dangerous situation, someone in distress, or violations of this policy, even if they seem inconsequential.

Participants asked to stop any harassing behavior are expected to comply immediately. If a participant engages in unacceptable behavior, the event organizers retain the right to take any actions to keep the conference a welcoming environment for all participants. This includes warning the offender or expulsion from the conference.

Program

Time	Session	Speaker(s)
9:00	Welcome from organizers	Selman Ipek and Pedro Pessoa
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10:00	Talk 2	Mohammad Abedi
10:20	Coffee Break and Group Photo	
11:00	Talk 3	Daniel Robbins
11:20	Talk 4	Pedro Pessoa
11:40	Talk 5	Nestor Caticha
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2:40	Coffee Break and Poster Session	
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3:50	Talk 8	Carlos C. Rodriguez
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11:00 - Talk 3: Daniel Robbins - Applications of the Moment Problem to Physics

11:20 - Talk 4: Pedro Pessoa - Neural Network Assisted Inference in Biological Dynamics

11:40 - Talk 5: Nestor Caticha - Prejudice and Discrimination in a Society of Neural Networks

1:40 - KEYNOTE: Ariel Caticha - Towards an Entropic Quantum Gravity

3:10 - Talk 6: John Kaminski - Tracking a Precessing Quantum Spin System

3:30 - Talk 7: Arthur Redgate - A Linear Model for X-ray Attenuation, Phase and Dark Field Imaging With Low-Coherence Sources

3:50 - Talk 8: Carlos C. Rodriguez - Robustness from Infinite Dimensional Ignorance

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How to Decide: An Information-Theoretic Approach for Exploring Partially Identified Complex Problems

Program slot: 9:20 - Talk 1

Presenter: Amos Golan

Co-authors: Jeffrey Perloff

Affiliation: Department of Economics, American University; Santa Fe Institute

Abstract

The available information is usually too complex and insufficient to ensure a unique solution in practically all modeling and inference problems. An information-theoretic (IT) methodology provides an alternative approach to finding solutions to partially identified (underdetermined) problems. In such models, we can identify only a solution set rather than point-identifying the parameters of interest, given our limited information. This solution set is usually an interval. Based on this incomplete information, how should one make a prediction, choose a treatment, or make other decisions to maximize welfare? The choice depends on whether the criterion is Bayesian, maximizing the minimum welfare (maximin), minimizing the maximum risk (Wald, 1944), information-theoretical, or other. A paper by Manski (2021) discusses the first three of these approaches, focusing on a statistical decision theory approach that minimizes the maximum risk or regret (minimax regret, or MMR). In this talk I will discuss an information-theoretical approach as an alternative. I compare it to other commonly used approaches. Using simulations for missing data and treatment problems in the social sciences, I will show that the IT performs the same or better than MMR. In additional simulations, IT dominates various other statistical decision functions. I will also present an empirical application. IT has an axiomatic underpinning and is computationally efficient.

Entropic Foundation of Finance

Program slot: 10:00 - Talk 2

Presenter: Mohammad Abedi

Co-authors: Daniel Bartolomeo

Affiliation: Department of Physics, University at Albany

Abstract

We develop an entropic framework for modeling stock dynamics and pricing European options. Rather than assuming a stochastic process from the outset, the model is derived through entropic inference, where probabilities are updated in the presence of incomplete information. A key ingredient is scale invariance: because investors care about returns rather than absolute prices, the formalism is constructed to remain invariant under rescaling of stock prices. This identifies the logarithm of price as the natural dynamical variable. Using the method of maximum relative entropy, we derive the short-time transition probability for the log price by imposing continuity of motion and a directional constraint related to expected return. The resulting dynamics reproduce geometric Brownian motion, and the probability density evolves according to a Fokker-Planck equation. Volatility enters through an entropic notion of time that serves as a natural clock for the dynamics. We then extend the framework to European option pricing. By incorporating the risk-neutral condition as an informational constraint, we derive the risk-neutral transition probability and use it to evaluate expected payoffs at maturity. Under constant volatility and risk-free rate, the formalism yields the Black-Scholes formulas for European call and put options, put-call parity, and the Black-Scholes-Merton partial differential equation. The main contribution is to provide an inferential foundation for familiar financial models: geometric Brownian motion and risk-neutral valuation emerge from entropic updating once the relevant information and symmetries are specified.

Applications of the moment problem to physics

Program slot: 11:00 - Talk 3

Presenter: Daniel Robbins

Affiliation: Department of Physics, University at Albany

Abstract

The moment problem in mathematics is easy to understand: given a list of real numbers, is it possible that they correspond to the moments of some probability distribution or density function (and is it unique, and can we construct it)? For single variable distributions on various domains the problem was essentially solved by mathematicians in the late nineteenth and early twentieth centuries. Because probabilities and densities appear throughout physics, the moment problem has an important and often underappreciated role to play. I will discuss a couple of historical applications and then describe some ongoing work where the moment problem can be used to constrain the space of two-dimensional conformal field theories with global symmetries.

Neural Network Assisted Inference in Biological Dynamics

Program slot: 11:20 - Talk 4

Presenter: Pedro Pessoa

Co-authors: Juan Andres Martinez; Vincent Vandenbroucke; Frank Delvigne; Steve Pressé

Affiliation: Center for Biological Physics, Arizona State University

Website: <https://pessoap.github.io/>

Abstract

Bayesian inference provides a rigorous probabilistic framework for analyzing data and learning underlying mechanisms while accounting for intrinsic stochasticity in the system as well as measurement uncertainty. However, many fields have not fully benefited from this framework. One key reason is that traditional Bayesian approaches rely on analytical likelihoods, which are not available for many problems of interest.

One such example is biological dynamics. Biological systems are fundamentally dynamical: cells grow, divide, and integrate information over time. At the same time, these processes are intrinsically stochastic, history-dependent (non-Markovian), and typically observed only through noisy, indirect measurements. These features make analytical likelihoods intractable and limit the applicability of standard inference tools.

In this talk, I show how neural networks can be used to approximate likelihoods directly from simulations, enabling Bayesian inference in otherwise intractable models. As a case study, we examine protein production across dividing cells, where standard approaches fail to account for division timing and molecular inheritance. By explicitly incorporating cell-division history, we find that gene activation events (particularly *glc3* expression in yeast measured via flow cytometry) are rarer and more transient than naive analyses suggest.

As time allows, I will also discuss related problems in gut microbiota and infection dynamics, where rare events across multiple biological scales similarly render analytical likelihoods unavailable, motivating a simulation-based Bayesian approach.

Prejudice and Discrimination in a Society of Neural Networks

Program slot: 11:40 - Talk 5

Presenter: Nestor Caticha

Affiliation: Instituto de Física, Universidade de São Paulo

Abstract

Opinion dynamics in agent-based models frequently use agents with typically no internal structure. We, however, consider agents that use an adaptive neural network to generate yes/no opinions about public issues. Agents exchange opinions and update a measure of distrust/trust towards other agents.

The neural networks (ideological sector) and the trust/distrust attribution (affinity sector) use algorithms obtained by a Maximum Entropy approximation to Bayesian learning, optimal in the limit of large dimensional issues and small societies.

Learning occurs primarily in the presence of dissonance, when the receiver agent disagrees and trusts the emitter, or reversely, agrees yet distrusts. This algorithm, despite being optimal in a small society, leads with high probability to a very polarized society; hence it is not optimal for group adaptation. Polarization becomes infrequent when the receivers use the optimal algorithm but change post-synaptic fields in order to be more tolerant towards the emitters. Next we consider societies which can be separated into two classes of agents. An agent is said to be prejudiced if they extend different tolerance fields toward in-group and out-group agents.

The phase diagram in the space of tolerance/prejudice field d vs. fraction of prejudiced agents f_d is constructed from three types of pair correlations combining opinions, distrust, and class. It shows several phases, among which is the emergence of collective discrimination. Two types of frustration in the sense of spin glasses (ideological and affective) are used to measure the balance of social relations, which depend not surprisingly on f_d and d , but more significantly on the size of the agenda.

In a polarized society, do out-group members hate each other because they disagree, or disagree because they hate each other? A society discussing a small agenda initially shows affinity polarization and then ideological polarization. Societies discussing large agendas take the opposite route.

Optimal learning in small groups, but not simpler algorithms, yields undesirable collective behavior. Our aim is to suggest measurements by social scientists of quantities that can be inferred from the macroscopic properties of societies.

Keynote – Towards an Entropic Quantum Gravity

Program slot: 1:40 - Keynote

Presenter: Ariel Caticha

Affiliation: Department of Physics, University at Albany

Abstract

The framework of Entropic Dynamics (ED) allows one to derive dynamical theories from well-established inference methods — probability, entropy, and information geometry. In particular, it allows us to reconstruct quantum theory as Hamilton-Killing flows on the cotangent bundle of a statistical manifold. These flows are such that they preserve the symplectic geometry and the information geometry of the statistical phase-space. They explain the linearity of quantum mechanics, the appearance of complex numbers, the quantum measurement problem, the nature of spin, and much more.

Our goal here is to deploy ED and information geometry to describe the kinematics of an Entropic Quantum Gravity (EQG). Our system consists of scalar matter and location fields [a field degree of freedom $\chi(x)$ is labeled by x , but it is located somewhere else at $y(x)$.] Most importantly, the ontology of the system does not include a spatial metric.

The Hamilton-Killing flow is described by a linear Schrödinger functional equation for the wave functional $\Psi[\chi, y]$, from which standard information geometry allows one to identify the emergent Riemannian metric of space. Thus, we have derived the quantum dynamics of a matter field χ interacting with location fields y , where the geometry of space is derived from the evolving wave functional $\Psi[\chi, y]$.

Some additional highlights are as follows. We impose invariance under spatial diffeomorphisms, as befits a relativistic theory of gravity. As one might expect, the superposition of a wave functional Ψ_1 with geometry g_1 with another wave functional Ψ_2 with geometry g_2 leads to a new quantum state Ψ_3 with its own definite geometry g_3 ; it does not lead to a superposition of geometries with their attendant indefinite causal structures. The possibility of a minimal length leading to a natural UV cutoff is pointed out. Here we only address the kinematic aspects of EQG and not the full foliation-invariant dynamics, which will be treated elsewhere.

Tracking a Precessing Quantum Spin System

Program slot: 3:10 - Talk 6

Presenter: John Kaminski

Affiliation: Department of Physics, University at Albany

Abstract

The parameters which govern the evolution of a quantum system can be determined by making measurements on an ensemble of identically-prepared systems. However, there are times when we are unable to recreate an experiment and want to measure physical parameters from just a single system. My work examines how physically-significant parameters associated with quantum systems can be measured via repeated coarse-grained measurements on a single copy of a quantum system. Specifically, I am looking to estimate the angular frequency of a quantum spin state undergoing Larmor precession using coarse-grained measurements on the particle's spin. I use a computer simulation of the spin and perform Bayesian analysis on the simulated data to extract the relevant information. The overarching goal of my research is to contribute towards a broader understanding of the classical-quantum correspondence. Similar work has been performed by others in the field showing that macrorealism, and thus macroscopic properties, emerge under coarse-grained measurements.

A Linear Model for X-ray Attenuation, Phase and Dark Field Imaging With Low-Coherence Sources

Program slot: 3:30 - Talk 7

Presenter: Arthur Redgate

Co-authors: Jonathan C. Petruccelli; Carolyn A. MacDonald

Affiliation: University at Albany, Department of Physics

Abstract

X-rays have been a cornerstone of medical and non-destructive materials imaging since their discovery by Röntgen in 1895. Though historically limited to absorption contrast, modern techniques can capture information about X-ray refraction and small-angle scattering due to microscopic density variations below the resolution of the detector. Recent advancements in computational imaging now provide access to these additional channels of information, through techniques known as differential phase contrast and dark-field imaging, respectively. For low-density materials, the differential phase contrast image can provide higher image contrast and highlights the edges of structural features, while the dark field image highlights fine structures that would be invisible to conventional x-ray imaging. While many methods for extracting these additional channels exist, most require high-coherence sources, limiting their applicability to synchrotron imaging or low-power microfocus laboratory sources. Low-coherence techniques using fine-pitched gratings exist, but require precise alignment and translations of the gratings, limiting their practical deployment. In this work, we discuss phase and intensity x-ray linearization with darkfield (PIXLD), an algorithm capable of retrieving all three channels of information that is compatible with low-coherence laboratory sources and has a very simple imaging setup that only requires that refraction angles are small. The system simply needs a coarse reference pattern capable of creating a suitable contrast pattern visible at the detector, in our case a stainless-steel wire mesh, and an actuator for shifting the pattern transverse to the beam direction. Beginning with a forward model that describes intensity modulations of reference pattern in the presence of a sample, a first-order Taylor expansion in refraction angle is employed to create a linearized form with a unique solution requiring only four shifts of the reference pattern. This model can also be iteratively extended to higher-order terms in the expansion, alleviating the requirement of small refraction angles. Simulation modeling has been employed to verify the computations.

Robustness from Infinite Dimensional Ignorance

Program slot: 3:50 - Talk 8

Presenter: Carlos C. Rodriguez

Affiliation: Department of Mathematics and Statistics, SUNY Albany

Website: <https://omega0.xyz/omega8008>

Abstract

In the framework of inductive inference – central to Entropic Dynamics and the physics as inference program – the assignment of probabilities naturally generates a hierarchical sequence of geometric abstraction spaces. We investigate the sequence $S_0 \rightarrow S_1 \rightarrow S_2$, where the base space S_0 represents the data manifold. Lifting this space yields S_1 , the statistical manifold of probability distributions (likelihoods) over S_0 . Inference on S_1 subsequently requires a prior, which resides in S_2 , the meta-manifold of probability distributions over S_1 .

A fundamental conceptual problem in this framework is the threat of infinite regress: does defining a state of maximum ignorance require an endless tower of priors ($S_n \rightarrow S_{n+1}$)? We show that under the plain

Information Metric, this abstraction sequence structurally terminates at level $S_0 \rightarrow S_2$. Specifically, we demonstrate that when S_1 is a non-parametric model, S_2 collapses into a space of finite Information Volume.

The existence of this finite volume allows for a proper, normalizable uniform prior on the manifold S_2 .

Consequently, the geometric state of absolute ignorance is fully specified without requiring further hierarchical ascent. We discuss the mathematical details of this termination, the role of the δ connection in the underlying geometry, and the implications for grounding physical law in a rigorous, non-regressive Theory of Ignorance.

The microscopic picture of black hole entropy

Program slot: 4:30 - Talk 9

Presenter: Oleg Lunin

Affiliation: Department of Physics, University at Albany

Abstract

This talk will review the information paradox associated with entropy of black holes and the resolution of this problem proposed by string theory. A particular emphasis will be placed on the geometric structure of black hole microstates and on recent results pertaining to testing the microscopic picture through scattering of gravitational waves.

Can We Unscramble Jaynes's Omelet? What Quantum No-Go Theorems Say About Probabilistic Reconstructions of Quantum Theory

Program slot: 4:50 - Talk 10

Presenter: Selman Ipek

Affiliation: Leibniz Universität Hannover

Website: <https://sites.google.com/view/selman-ipek/>

Abstract

In the standard formulation of quantum mechanics, the probabilistic aspects of the theory are introduced in a way that is inseparable from the process of measurement. This is unsatisfactory, since the notion of measurement is itself ambiguous. Ontological models provide a framework for representing quantum measurement statistics using probability distributions over underlying physical states. Although such models ground quantum probabilities in ordinary probability theory, they are often silent about what, precisely, constitutes a measurement in the first place.

In this talk, I will first review some of the most significant no-go theorems in the foundations of quantum mechanics and discuss which classes of ontological models they do and do not rule out. I will then turn to the Entropic Dynamics approach to quantum mechanics, which can be understood as a realist, psi-epistemic ontological model. I will argue that this framework not only evades the usual no-go theorems, but also offers a resolution of the measurement problem.

The Statistical Mechanics of Motion

Program slot: 5:10 - Talk 11

Presenter: Kevin Knuth

Affiliation: Department of Physics, University at Albany

Website: <https://www.albany.edu/physics/faculty/kevin-knuth>

Abstract

Given the prominence of the concept of motion in the physical world, it is surprising that in the last two thousand years, few concepts of physics have demanded more attention, and have been so little advanced by experimentation, than motion. Physics had seemingly escaped from the doctrine of Parmenides, which held that motion was an illusion, and the nine paradoxes of Zeno of Elea, through the quantification of motion by Galileo's work on kinematics and the causal connection of motion to forces by Newton's work on dynamics. Motion took its central place in physics, and was seemingly straightforward until Einstein clarified the relationship between space and time, which again made motion mysterious - specifically the fact that the speed of light is the maximum speed, and that it is impossible for massive particles to attain it. For most, the strangeness of motion is easily ignored as it occurs in a normally inaccessible regime of the highest speeds. However, around 1930, Gregory Breit and Erwin Schrödinger showed that the eigenvalues of the velocity of a particle described by wavepacket solutions to the Dirac equation are simply $\pm c$, the speed of light. This led Schrödinger to coin the term Zitterbewegung, which is German for "trembling motion," where all particles of matter (fermions) zig-zag back and forth at only the speed of light. The result is that any finite speed less than c , including the state of rest, only makes sense as a long-term average that can be thought of as a drift velocity. As a result, the concept of motion, again, is fraught with difficulties. Relativity suggests that massive particles cannot move at the speed of light, and relativistic quantum mechanics suggests that massive particles can only move at the speed of light. We consider the problem of Zitterbewegung for a particle and two observers, and show that while, on the average, the particle has an instantaneous velocity of the speed of light, the speeds that the observers measure is consistent with the relativistic velocity addition rule, such that massive particles cannot be observed to go the speed of light. The result is a statistical mechanics of motion, in which the maximum entropy state is the state of rest, and the act of adding energy to the particle results in a decrease of entropy that is accompanied by motion.